Lecture 10: Generative Models for Life Sciences

Controlling Diffusion Models for Protein Design

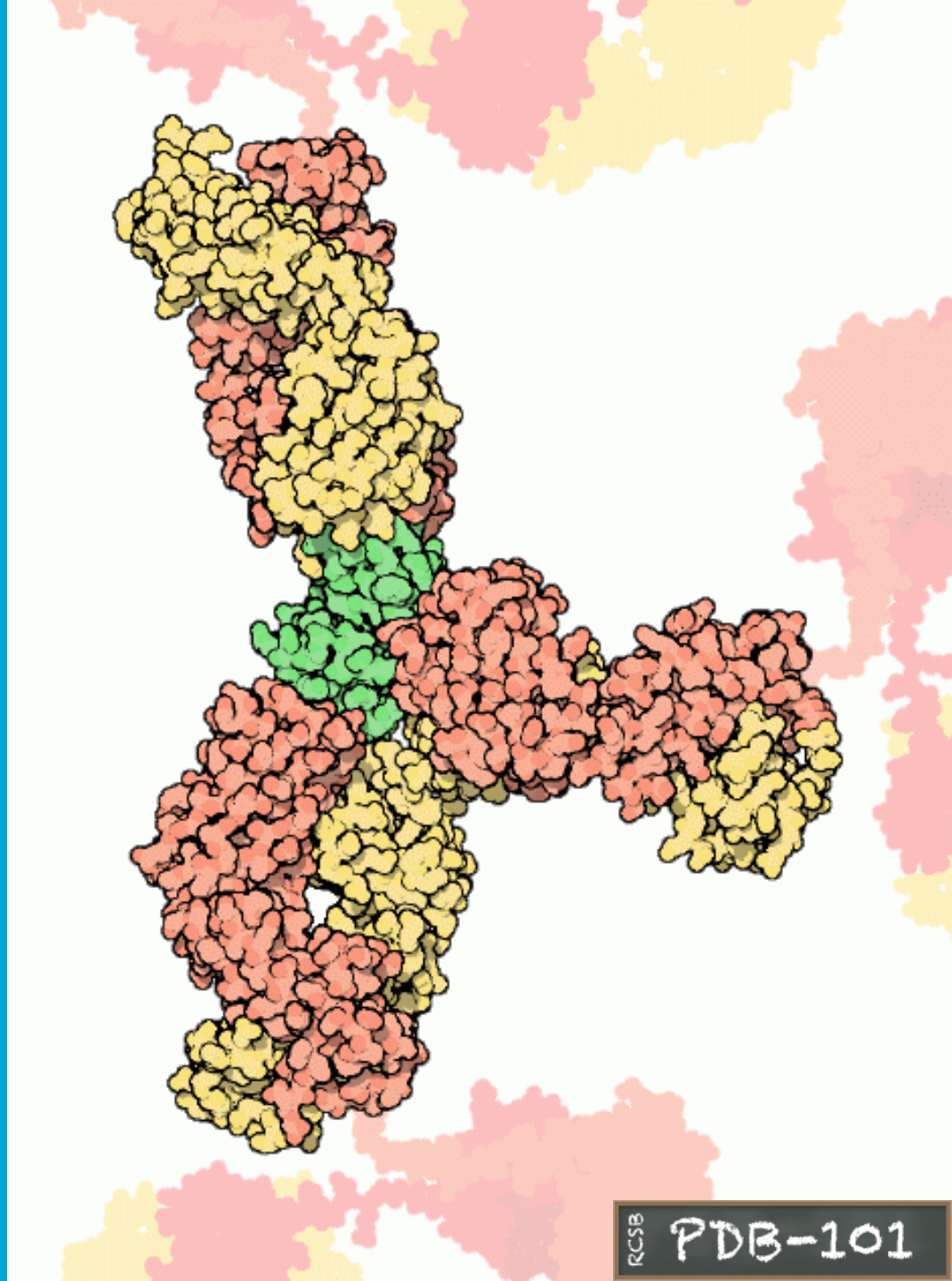
DSAIT4030 - Generative Modeling 02-06-26

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Part I – From diffusion to controlled generation

In the first 45 mins we will…

- Motivate the need for **controllable diffusion**
- Learn mechanisms of **conditioning & guidance** in diffusion models
- Introduce **representations for protein** generation

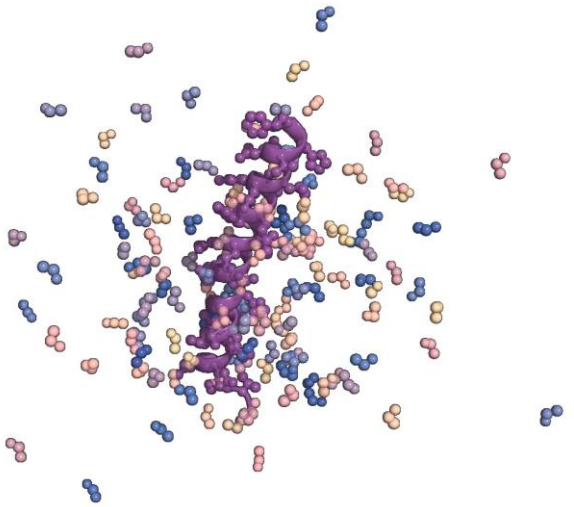
Why controllable diffusion models?

- Well-known success in **image and video generation**
- Example: *Transform images into visuals with a few descriptive words*



Source: NVIDIA Research

- But also, revolutionized the fields of **protein and small molecule design**
- Example: *Generate novel proteins that bind to target molecules*



Source: RFDiffusion

Core question:

How to sample useful data points, not just plausible ones?

Recap: Diffusion as iterative denoising

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = N\left(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

iteratively add noise on **forward** process

$q(\mathbf{x}_0)$ is unknown

$q(\mathbf{x}_T|\mathbf{x}_{T-1}) \approx N(0,1)$

Data
$(t = 0)$



Noise
$(t = T)$

learn to remove this noise on **reverse** process

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = N\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)\right)$$

$p(\mathbf{x}_T) = N(0,1)$
easy to sample from

$$L_{simple} = \mathbb{E}_{\mathbf{x}_0,t,\boldsymbol{\epsilon}} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2$$

Source: Ho et al. (2020) - Denoising diffusion probabilistic models

Recap: DDPM training and sampling

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

predicted noise

$\mathbf{x}_t$, noised sample at timestep t

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ → sample from prior
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \mathbf{z}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

predicted noise

add noise back for stochasticity

Source: Ho et al. (2020) - Denoising diffusion probabilistic models

Controlling the generation process

Unconditional

Sample from $p(\mathbf{x})$

- Broad data distribution
- e.g., images of animals



$p(\mathbf{x})$





Conditional

Sample from $p(\mathbf{x} \mid \mathbf{y})$

- Target a specific class
- e.g., cat images



$p(\mathbf{x} \mid \mathbf{y} = \text{"cat"})$





Controlling the generation process

Learn a **conditional generative model** $p(\mathbf{x} \mid \mathbf{y})$ by training on data pairs $(\mathbf{x}, \mathbf{y})$ (e.g., images and captions)



Conditioning vs guidance

Often used interchangeably in literature; both model conditional $p(\mathbf{x} \mid \mathbf{y})$; differences:

Conditioning

Changes in the target distribution

$$p(\mathbf{x}) \rightarrow p(\mathbf{x} \mid \mathbf{y})$$

The condition is part of the model input or probabilistic formulation

What should be generated given context?
e.g., context-given inpainting

Guidance

Changes in the sampling trajectory

$$\mathbf{x}_t \leftarrow \mathbf{x}_t + \alpha \nabla_{\mathbf{x}} f(\mathbf{x}_t)$$

An external objective steer generation during sampling towards desired regions

Where should the sample move to?
e.g., gradient-based guidance

Classifier guidance

Sampling from a conditional generative model:

forward model (classifier)

$$p(\mathbf{x} \mid \mathbf{y}) = \frac{p(\mathbf{x})p(\mathbf{y} \mid \mathbf{x})}{p(\mathbf{y})}$$



However, we need an **estimation of the score function**:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{y}) = \nabla_{\mathbf{x}} \log \frac{p(\mathbf{x})p(\mathbf{y} \mid \mathbf{x})}{p(\mathbf{y})}$$

$$= \nabla_{\mathbf{x}} \log p(\mathbf{x}) + \nabla_{\mathbf{x}} \log p(\mathbf{y} \mid \mathbf{x}) - \nabla_{\mathbf{x}} \log p(\mathbf{y})$$

0

$$= \nabla_{\mathbf{x}} \log p(\mathbf{x}) + \nabla_{\mathbf{x}} \log p(\mathbf{y} \mid \mathbf{x})$$

unconditional score $\approx s_\theta(\mathbf{x})$

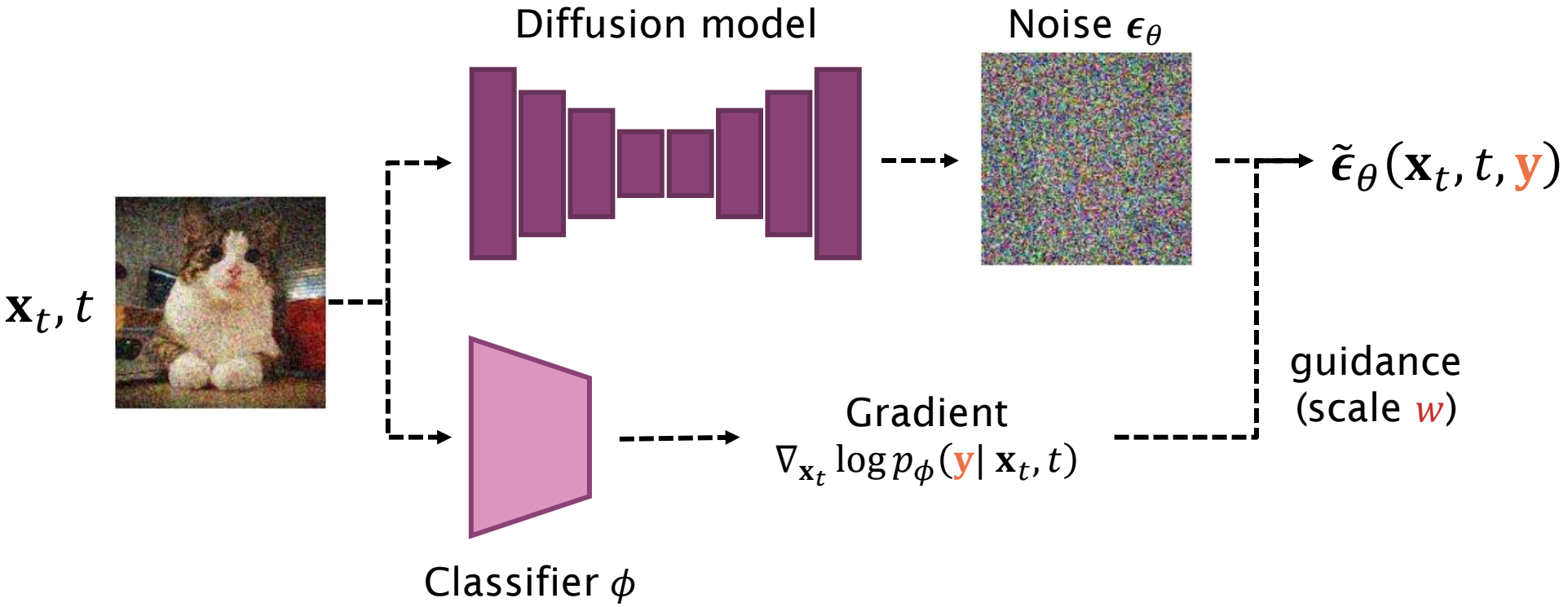
classifier gradient



Different forward models for the same score model

Source: Dhariwal & Nichol (2021) - Diffusion Models Beat GANs on Image Synthesis

Classifier guidance on timestep t



$$\tilde{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{y}) = \epsilon_\theta(\mathbf{x}_t, t) + w\sigma_t \nabla_{\mathbf{x}_t} \log p_\phi(\mathbf{y}|\, \mathbf{x}_t, t)$$

Classifier-free guidance (CFG)

Classifier guidance **requires training a forward model (classifier)** for all noisy versions of the data samples

Can we do better? → **Classifier-free guidance!**

Train both conditional + unconditional diffusion model together (by randomly dropping the condition $\mathbf{y}$) → **implicitly learn the classifier**

$$\nabla_{\mathbf{x}} \log p(\mathbf{y} \mid \mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{y}) - \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

implicit classifier! **conditional diffusion model** **unconditional diffusion model**

Source: Ho & Salimans (2021) - Classifier-Free Diffusion Guidance

CFG modified score

Combining the two models with a guidance strength w

$$\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{y}) = \nabla_{\mathbf{x}} \log p(\mathbf{x}) + w \nabla_{\mathbf{x}} \log p(\mathbf{y} \mid \mathbf{x}) \longrightarrow \text{classifier guidance}$$

$$= \nabla_{\mathbf{x}} \log p(\mathbf{x}) + w[\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{y}) - \nabla_{\mathbf{x}} \log p(\mathbf{x})]$$

Predicted noise at timestep t:

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t, \mathbf{y}) = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset) + w[\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{y}) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset)]$$

where $\emptyset$ means empty conditioning (dropped during training)

CFG sits at the boundary!

- **Conditioning**: model learns to denoise under $\mathbf{y}$ → approximate $p(\mathbf{x} \mid \mathbf{y})$
- **Guidance**: at sampling, amplify conditional direction with w → change trajectory

Source: [Ho & Salimans (2021) - Classifier-Free Diffusion Guidance]

CFG sits at the boundary

$$\tilde{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{y}) = \epsilon_\theta(\mathbf{x}_t, t, \emptyset) + w[\epsilon_\theta(\mathbf{x}_t, t, \mathbf{y}) - \epsilon_\theta(\mathbf{x}_t, t, \emptyset)]$$

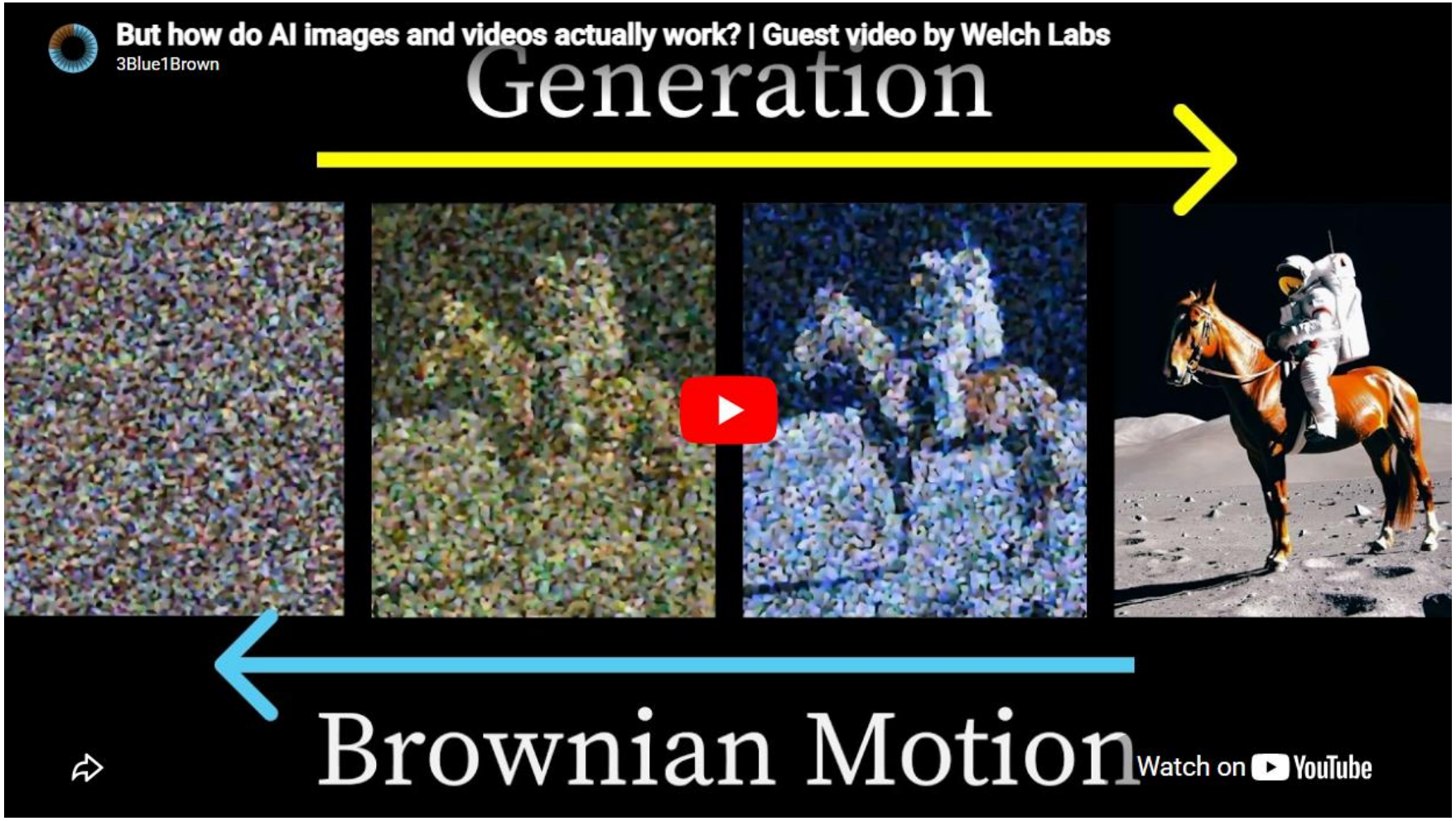
$w = 0$: unconditional sampling; $w = 1$: conditional sampling; $w > 1$: stronger guidance



w

Source: Ho & Salimans (2021) - Classifier-Free Diffusion Guidance

Video on conditional generation

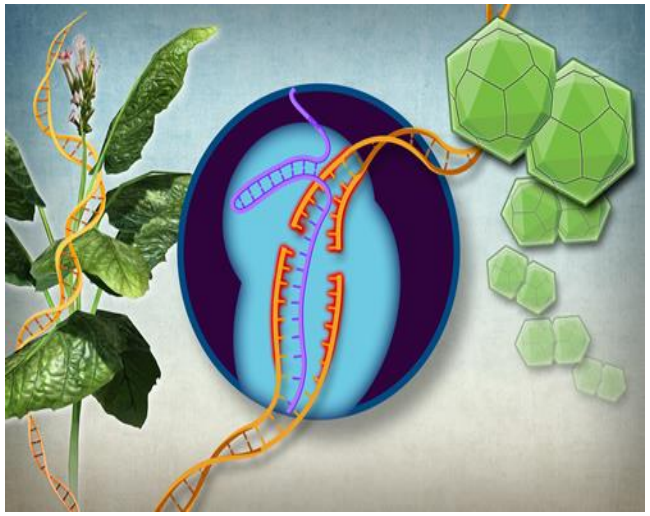


Wait a second…

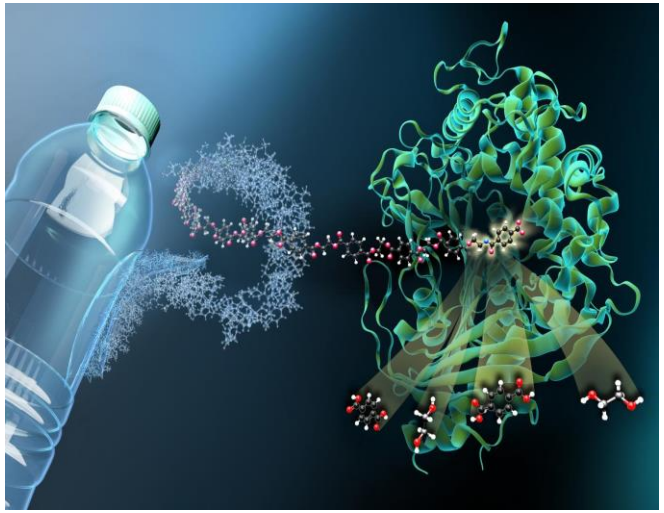
But why are proteins an interesting use case for generative modeling?

Diverse protein applications

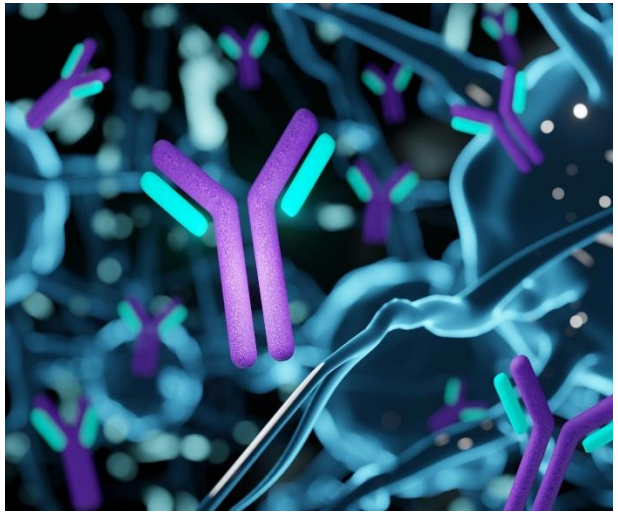
- Nature has **evolved** proteins to perform necessary functions for life
- Humans have **engineered** proteins for different applications



Genome editing
E.g., CRISPR for plants



Bioprocessing
E.g., Plastic degradation



Drug discovery
E.g., Biotherapeutics

Images from: NIH Genome News & Genetics Engineering and Biotechnology News

Proteins come in different modalities!

Determines

Determines

Sequence

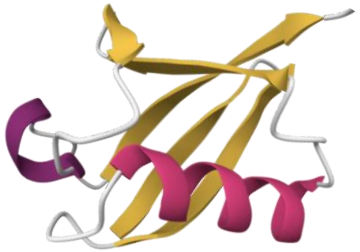
$\mathbf{x}$ = amino acid tokens (vocabulary of ~20)

MQIFVKTLTGKTITLEVEPSDTIE
NVKAKIQDKEGIPPDQQRLIFAG
KQLEDGRTLSDYNIQKESTLH…

- Sequence alignments
- Amino acid properties
- Protein language model embeddings

Structure

$\mathbf{x}$ = 3D atom coords, local frames or angles



- 3D atom coordinates (Cα, backbone, or full), angles
- Distance maps (Cα-Cα)
- Graphs (threshold on maps)

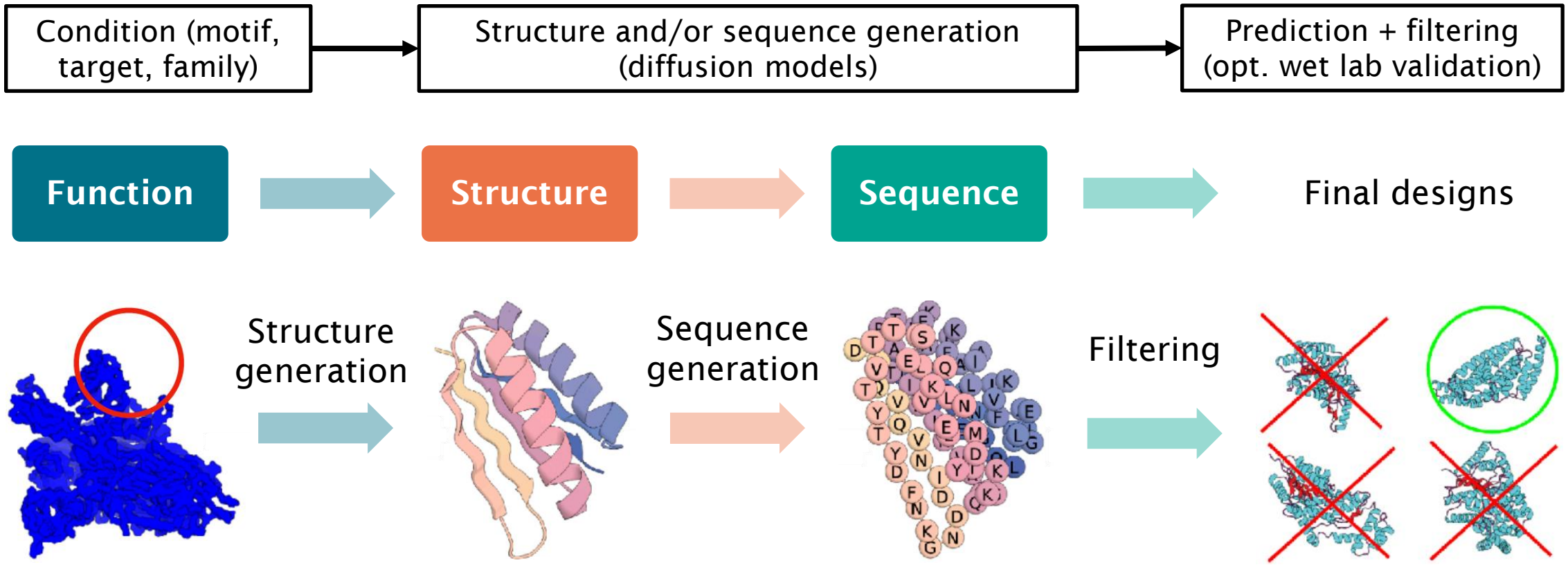
Function

$\mathbf{x}$ = family, interactions, cellular location, etc.

Ubiquitin, small regulatory protein that tags obsolete proteins for destruction

- Text (natural language)
- Labels (discrete or continuous)

Generative protein design pipeline



Credits: [Jason Yim (2025) – MIT Introduction to Flow Matching and Diffusion Models]

Time for a break!

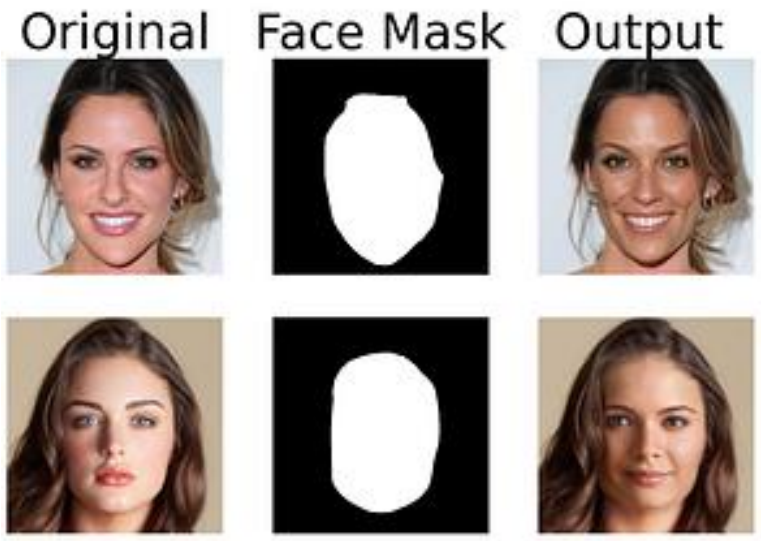
Part II – Controlling protein design

In the second 45 mins we will show three case studies…

- **Motif scaffolding** = *geometric* conditioning
- **Antibody co-design** = *modality + interaction* conditioning
- **Latent diffusion** = *latent semantic* guidance

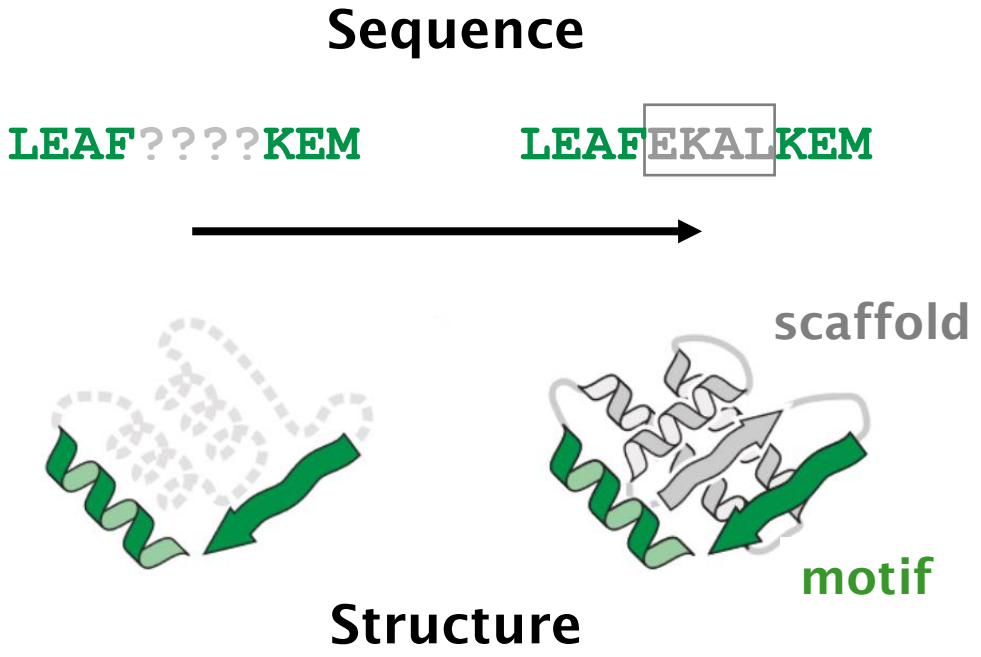
The motif-scaffolding problem

Similar to image inpainting problem



Motif → functional site of the protein

Scaffold → structure that supports a motif



Source: Wang et al. (2022) - Scaffolding protein functional sites using deep learning

Why motifs matter

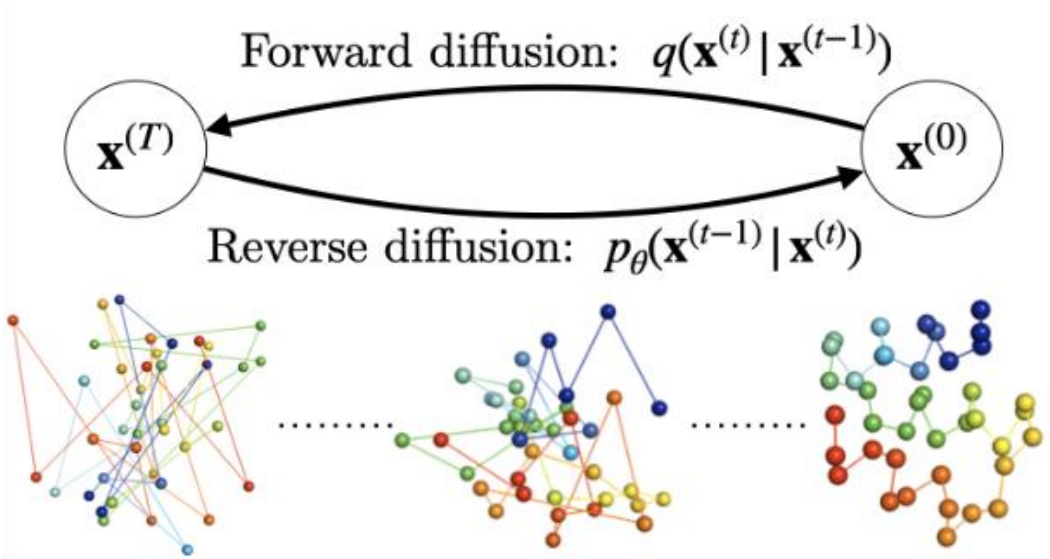
- In many protein design cases, we are not interested in modifying the motif to **maintain the protein function**
- In turn, we want to **optimize the scaffold** to get a protein with better properties (e.g., more stable)
- **Conditional generation problem** → given a functional motif, generate a protein scaffold that supports it

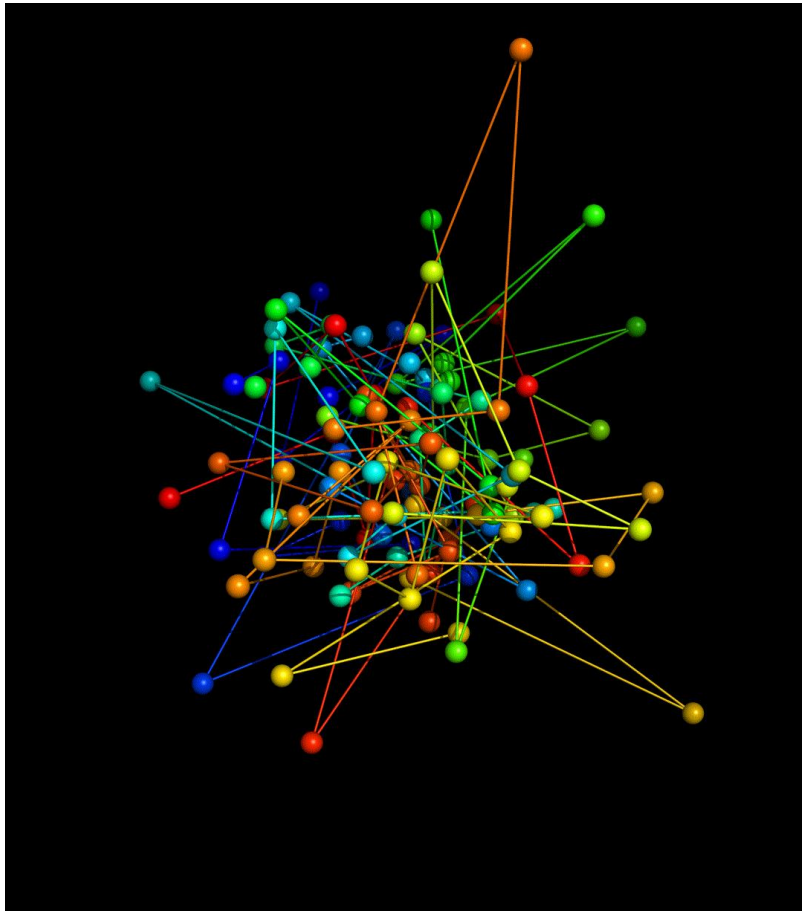
$$p(\mathbf{x} = \text{scaffold} \mid \mathbf{y} = \text{motif})$$

- Example of motif: enzyme active sites

Unconditional protein structure design

Gaussian diffusion (standard DDPM) on 3D coordinates for **Cα atoms** (one per amino acid)





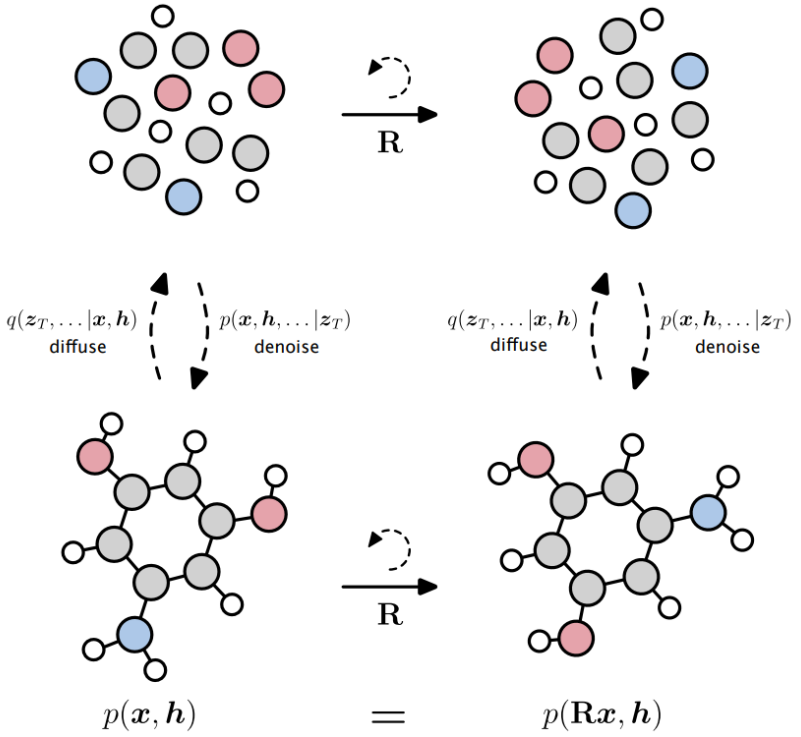
Source: [Trippe et al. (2023) - Diffusion probabilistic modeling of protein backbones in 3D for the motif-scaffolding problem]

Structure diffusion needs geometric models

Desired property → **Equivariance**

- **Translations, rotations, and reflections** applied to the input correspond to the same transformations at the output

$$\mathbf{R}f(\boldsymbol{x}) = f(\mathbf{R}\boldsymbol{x})$$



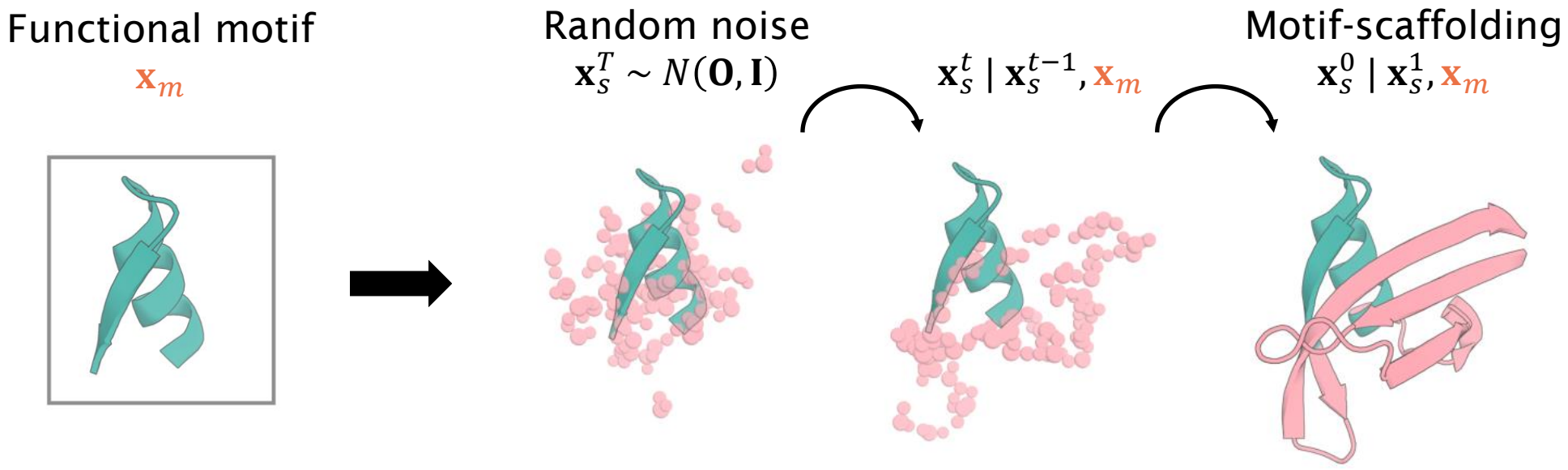
Denoiser → **Equivariant GNN**

- Protein as a **graph**: nodes are Cα atoms, edges represent spatial proximity (Cα-Cα)
- EGNN messages use **pairwise distances**: $\mathbf{m}_{ij} = \mathrm{MLP}\left(\mathbf{h}_i, \mathbf{h}_j, \left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2\right)$
- Coordinate updates use **relative positions**: $\mathbf{x}_i \leftarrow \mathbf{x}_i + \sum_{j \neq i}(\mathbf{x}_i - \mathbf{x}_j)\mathrm{MLP}(\mathbf{m}_{ij})$

Source: [Hoogeboom et al. (2022) - Equivariant Diffusion for Molecule Generation in 3D]

Motif-conditioned protein structure design

- **Motif is fixed** during generation
- Random noise $\mathbf{x}_s^T$ + motif $\mathbf{x}_m$ → input to **RFdiffusion**
- Iteratively refines that noise until a final protein structure is designed



Source: [Watson et al. (2023) - De novo design of protein structure and function with RFdiffusion]

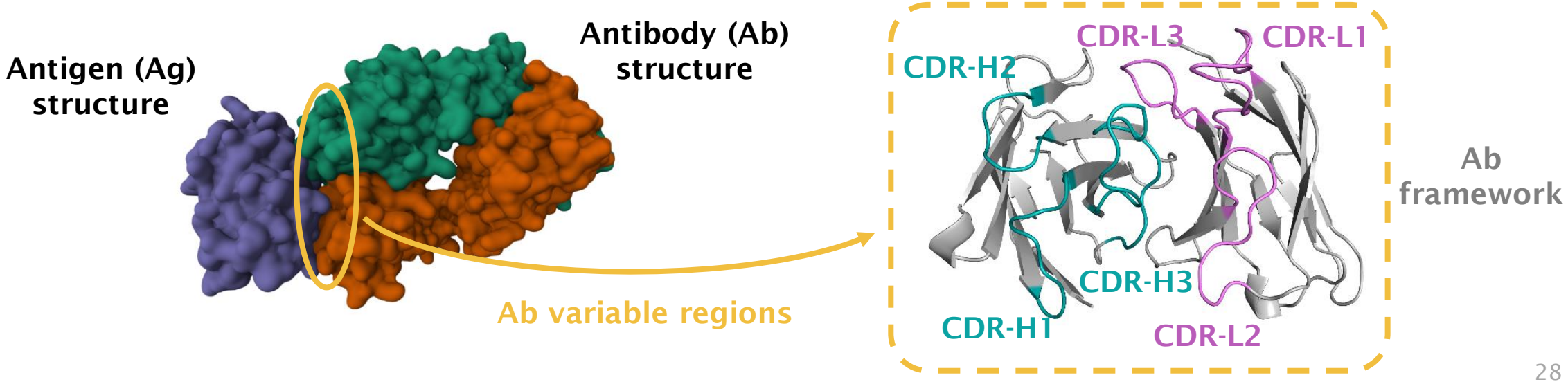
We have seen an example of intra-protein conditional backbone design…

How can we extrapolate this to protein-protein interaction design?

The antibody design problem

- **Antibodies** are proteins produced by the immune system in response to **antigens**
- **CDR loops** determine interaction → affinity and specificity
- **Conditional generation problem** → generate antibody CDRs that bind to antigen, given antibody framework

$$p(\mathbf{x} = \text{antibody CDRs} \mid \mathbf{y} = \text{antigen, antibody framework})$$



Diffusion for antibody design

- For antibody design, diffusion models can **generate CDR sequences and structures**

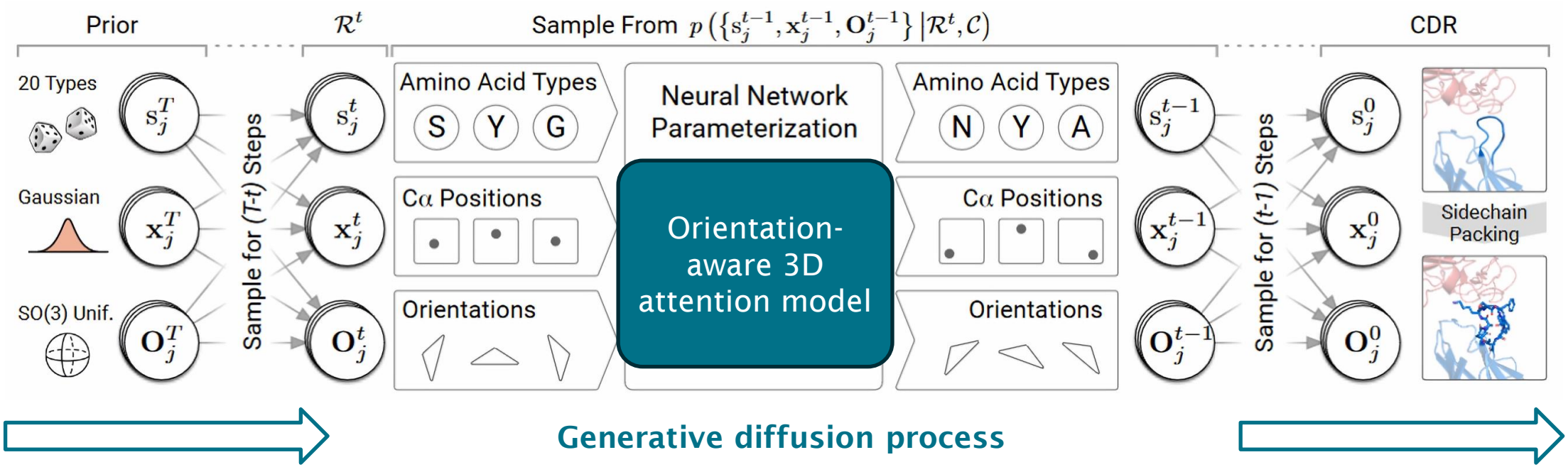
- **Multimodal**:
  - Sequence (discrete) → amino acid types
  - Backbone geometry (continuous) → Cα atom coordinates + local orientations

- **Conditioning signals**:
  - Three modalities of bound antigen + antibody framework (around CDRs)

Source: Luo et al. (2022) - Antigen-specific antibody design and optimization with diffusion-based generative models

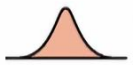
Diffusion for antibody design

- Three modalities → three **interacting diffusion processes**
- **Common denoiser** for sequence and structure → equivariant



Source: [Luo et al. (2022) - Antigen-specific antibody design and optimization with diffusion-based generative models]

Three interacting diffusion processes

Modality	Forward diffusion process ($t = 0, \ldots, T$) Distributions q	Prior distribution (sampling at $t = T$)	
Amino acid types	Multinomial distribution: $q(s_j^t \mid s_j^0) = \text{Multinomial}\left(\bar{\alpha}^t \cdot \text{onehot}(s_j^0) + \frac{1-\bar{\alpha}^t}{20} \cdot \mathbf{1}\right)$	Uniform distribution over 20 classes	20 Types 
Cα 3D coordinates	Normal distribution: $q(\mathbf{x}_j^t \mid \mathbf{x}_j^0) = N\left(\mathbf{x}_j^t \mid \sqrt{\bar{\alpha}^t} \cdot \mathbf{x}_j^0, (1-\bar{\alpha}^t)\mathbf{I}\right)$	Standard normal distribution	Gaussian 
Orientations in SO(3)	Isotropic Gaussian distribution in SO(3): $q(\mathbf{O}_j^t \mid \mathbf{O}_j^0) = IG_{\text{SO(3)}}\left(\mathbf{O}_j^t \mid \text{ScaleRot}\left(\sqrt{\bar{\alpha}^t} \cdot \mathbf{O}_j^0\right), 1-\bar{\alpha}^t\right)$	Uniform distribution over SO(3)	SO(3) Unif. 

Conditional sampling:

$$p\left(\left\{s_j^{t-1}, \mathbf{x}_j^{t-1}, \mathbf{O}_j^{t-1}\right\} \mid R^t, C\right)$$

$$p(\mathbf{x} = \text{CDR}^{t-1} \mid \mathbf{y} = \text{CDR}^t, \text{Ag}, \text{Ab framework})$$

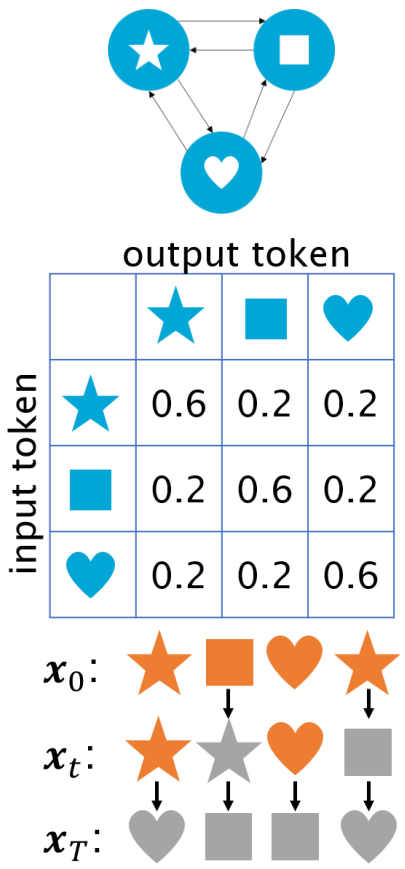
Source: Luo et al. (2022) - Antigen-specific antibody design and optimization with diffusion-based generative models

Discrete diffusion for sequences

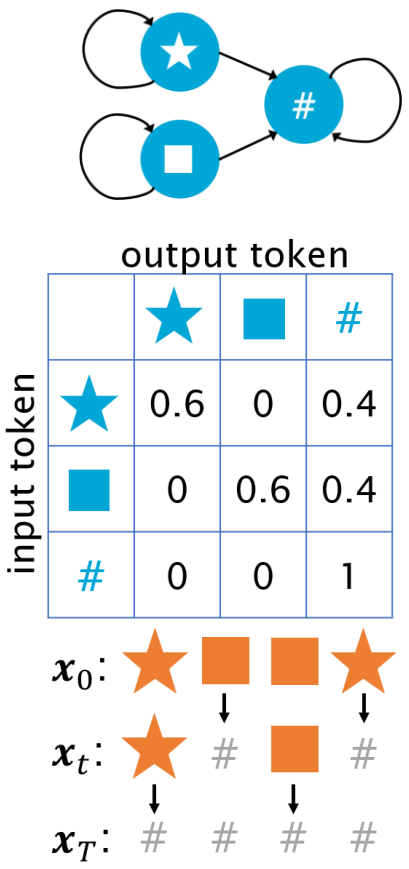
- Instead of Gaussian noise, use **token corruption** through Multinomial diffusion
- **Noise** → probability of transitioning from one category to another (uniform) or to a or to a mask [#] state (absorbing)
- Defined by **transition matrices** $\mathbf{Q}_t$
- **Forward process**:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; \boldsymbol{p} = \mathbf{x}_{t-1}\mathbf{Q}_t)$$

Uniform noise



Absorbing noise



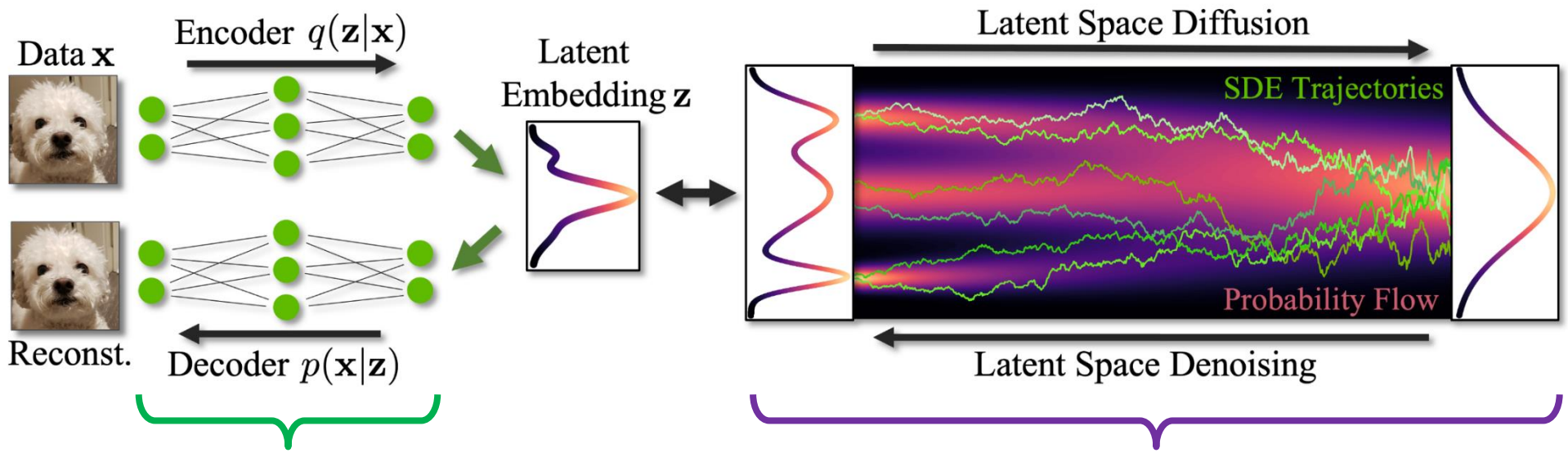
Sources: Austin et al. (2021) - Structured denoising diffusion models in discrete state-spaces
Alamdari et al. (2023) - Protein generation with evolutionary diffusion: sequence is all you need

Challenges increase when diffusing several modalities in the data space…

Is there a better way to diffuse discrete or complex data?

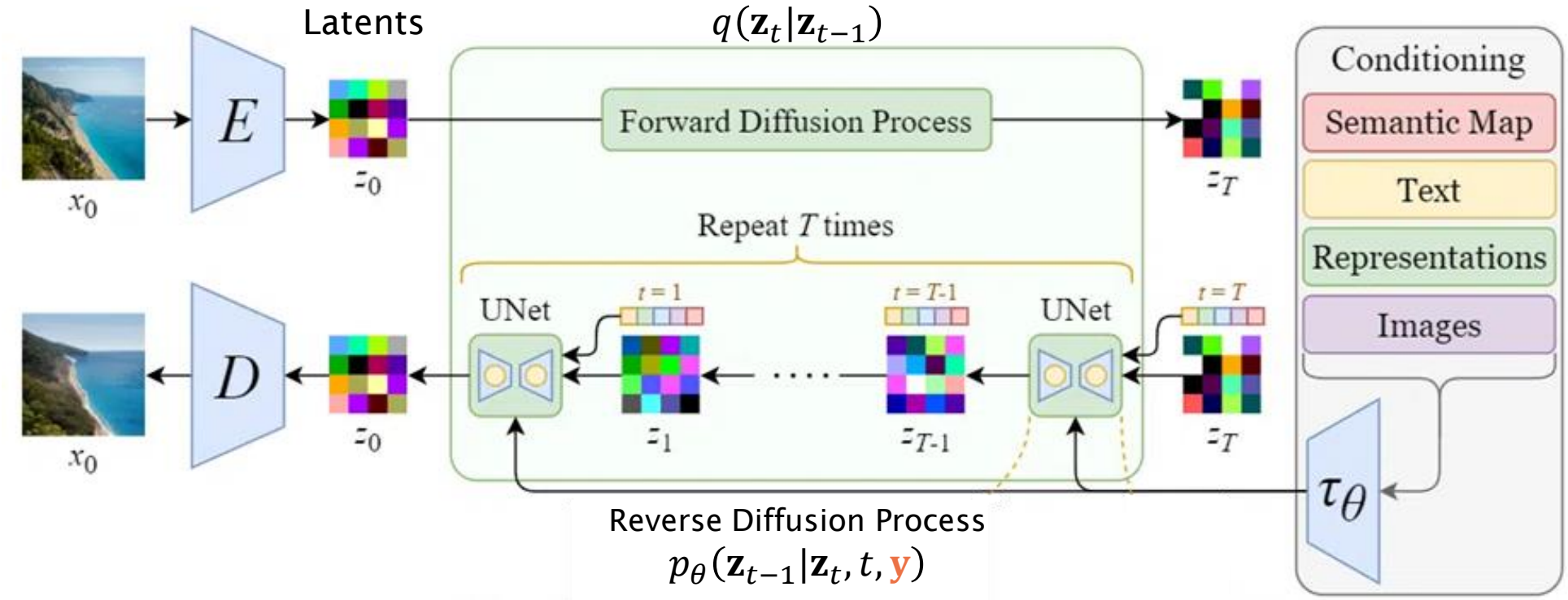
Latent diffusion!

- **Encoder** to map data $\mathbf{x}$ into (lower-dim) latent embeddings $\mathbf{z}$
- Forward + reverse Gaussian **diffusion** on the latent space
- **Decoder** to reconstruct data $\tilde{\mathbf{x}}$ from denoised latent embeddings $\tilde{\mathbf{z}}$



Source: [Latent Diffusion Models - NeurIPS 2023 Tutorial]

Guidance in latent diffusion



- Classifier-free guidance → on-off switch for the condition $\mathbf{y}$
- Conditioning is added through concatenation or cross-attention

Source: [Rombach et al. (2022) - High-Resolution Image Synthesis with Latent Diffusion Models]

Latent diffusion for protein sequences

- Structural data is complex (geometry, preserve equivariance) and highly scarce
  → protein **sequence data** is abundant and easier to manage

- Discrete diffusion requires **sampling from a Multinomial** distribution

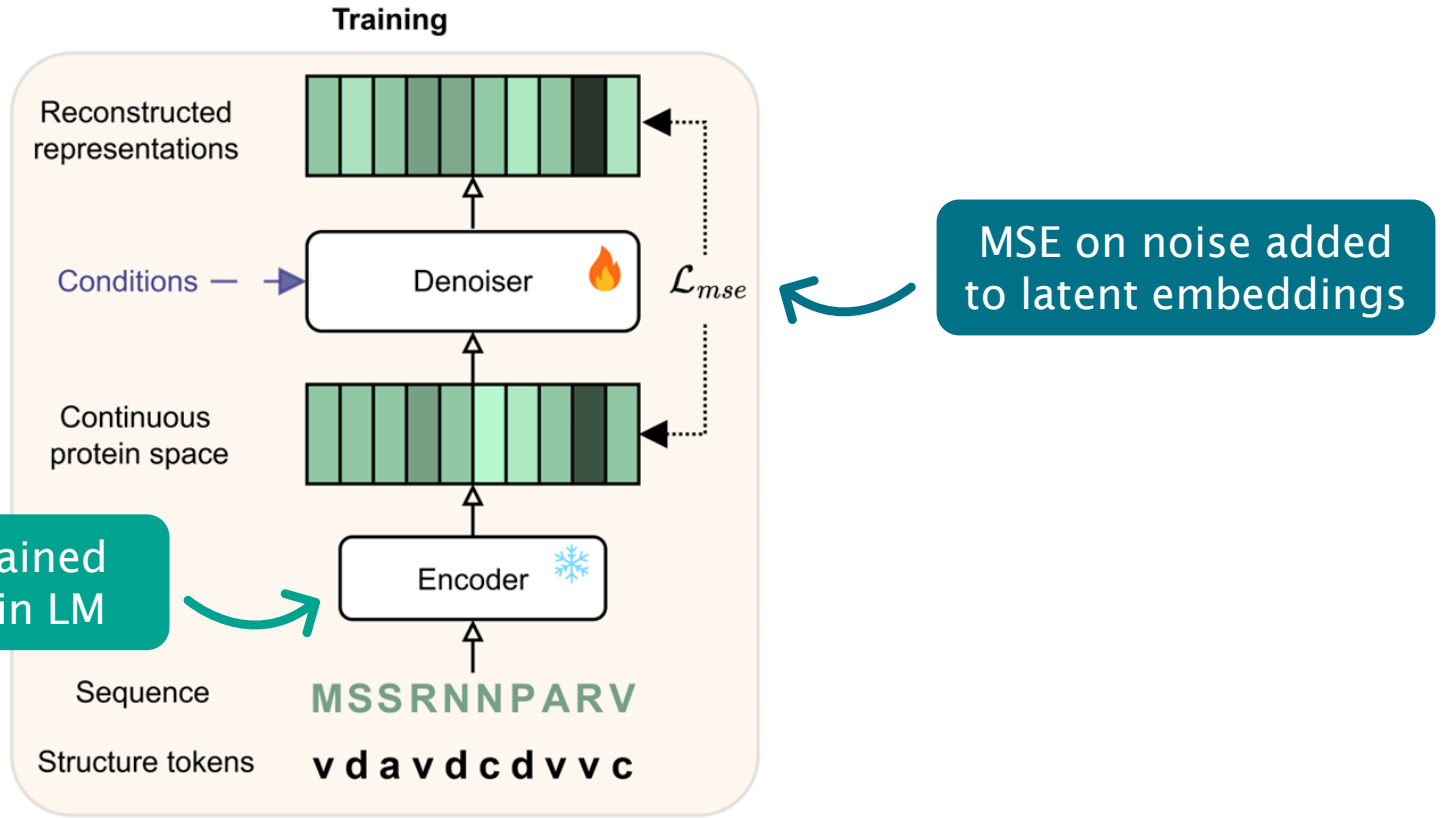
- Can we use **continuous representations for protein sequence design**?
  → DiMA: diffusion on protein language model embeddings (e.g., ESM-2)

- **Conditional generation problem** → generate protein sequence embeddings conditioned on motif, family, fold, text, etc.

$$p(\mathbf{x} = \text{protein LM embeddings} \mid \mathbf{y} = \text{motif, fold, text})$$

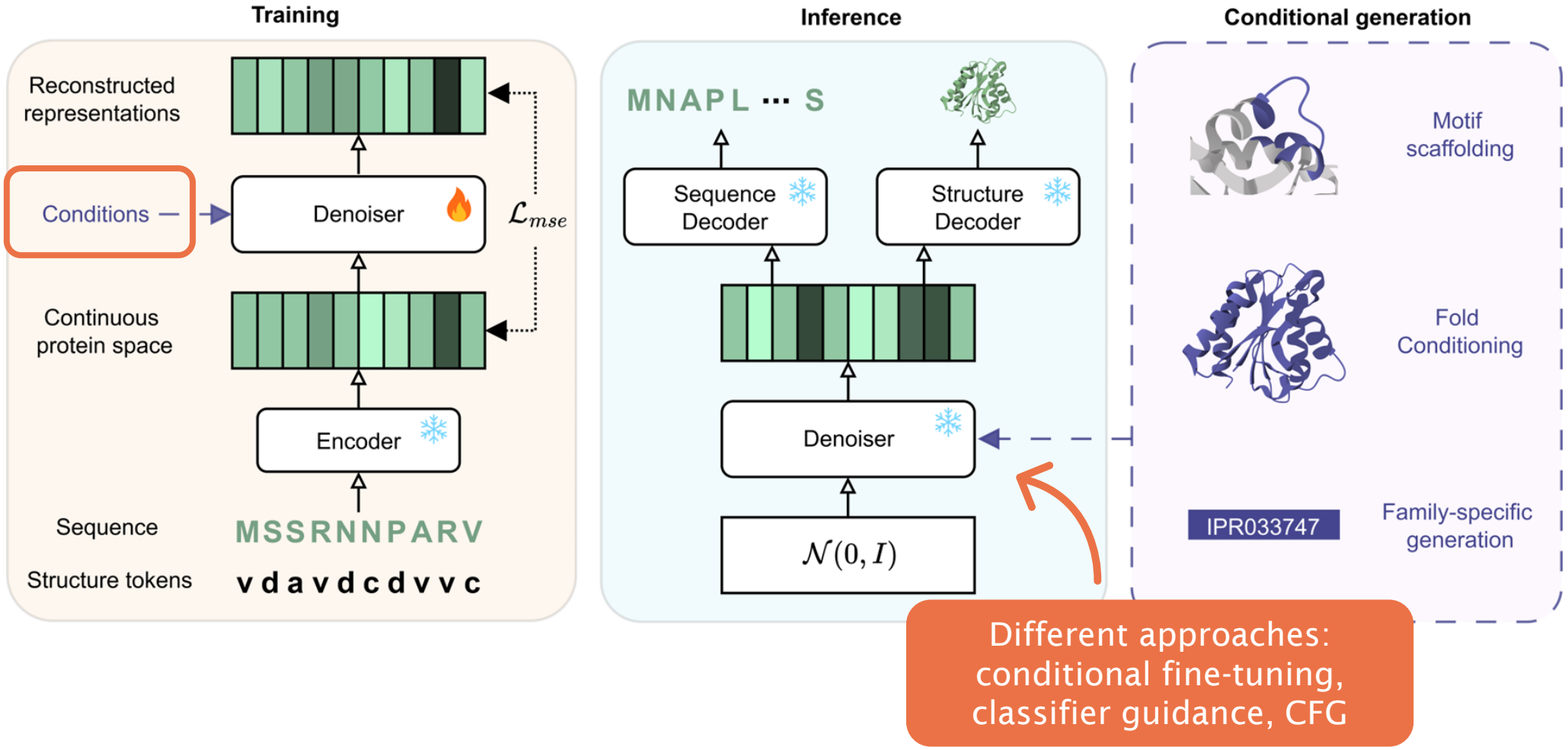
Source: Meshchaninov et al. (2025) - Diffusion on language model encodings for protein sequence generation

Diffusing protein language model embeddings



Source: [Meshchaninov et al. (2025) - Diffusion on language model encodings for protein sequence generation]

Diffusing protein language model embeddings



Source: [Meshchaninov et al. (2025) - Diffusion on language model encodings for protein sequence generation]

The three cases put together

Example	Conditioning	Guidance	Diffusion space
Motif scaffolding	Motif geometry	Stability	3D structure
Antibody design	Antigen + antibody framework	Affinity / developability	Sequence + geometry
Latent sequence design	Motif / fold	Family	Protein LM embeddings

Control becomes increasingly abstract: **geometry → interaction → latent semantics**

Lecture recap

In this ~90 min, we have detailed:

- Why controlled diffusion matters
- How conditioning / guidance is implemented
- Why protein design is a compelling domain for generative modeling
- What changes when data are structured / geometric / discrete
- Three case studies for protein generation

Questions?

Interested in these research topics?

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